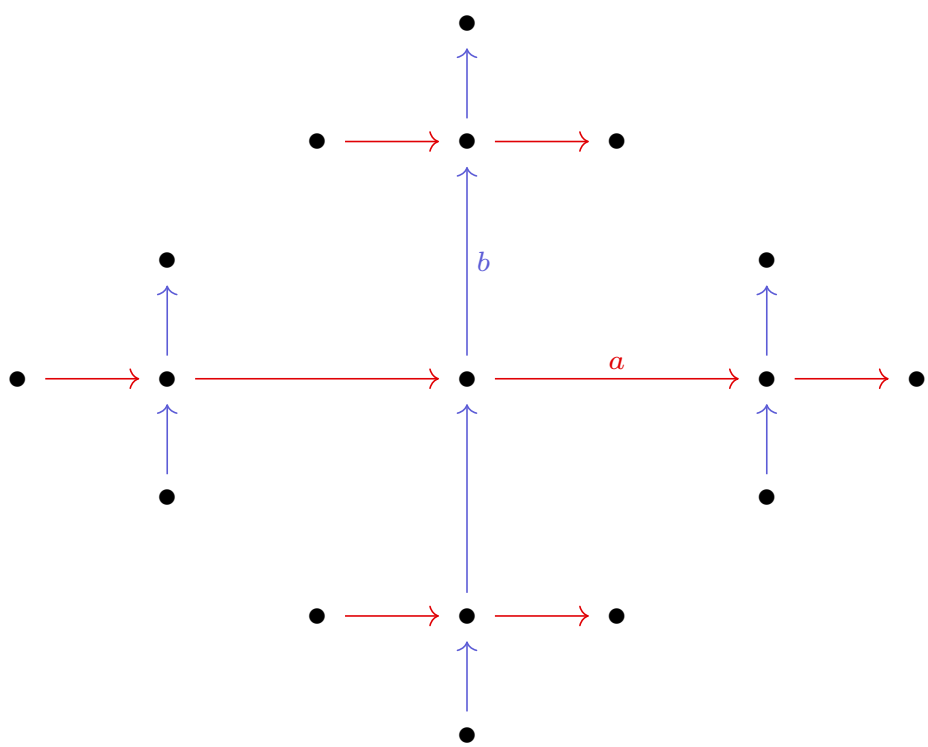
When Groups Act Out: The Ping-Pong Lemma & Serre's Theorem

Brandon Papandrea

Abstract

Free groups have played an important role in the history of both finite and infinite group theory since their inception, hiding their unique and complicated structure behind a very simple presentation. Two notable examples of this come to mind: there is no finite algorithm that can determine if an arbitary group presentation is that of a free group, and the question of if the subgroups of a free group are free. Nielsen and Schreier's original proof of the latter is among the most complicated math proofs of the 20th century.

Thankfully, there are many ways to tell if a group is free simply by studying the way it acts on a space. The Ping-Pong Lemma provides both a necessary and sufficient condition for determining the freeness of a group, assuming some basic properties of a group action, while Serre's Theorem provides a classification of free groups based on how they act on trees, proving the freeness of subgroups in the processs. This paper attempts to provide a singular, comprehensive overview of each result with full proofs and explanations for each.



Contents

Introduction **2**

1 The Ping Pong Lemma **3**
1.1 Definitions . . . 3
1.1.1 The Universal Property . . . 3
1.1.2 The Rank of Subgroups of F_2 . . . 6
1.2 The Ping-Pong Lemma . . . 7
1.2.1 Statements and Proofs . . . 7
1.2.2 An Application: $SL_2(\mathbb{Z})$ . . . 9

2 Serre's Theorem **11**
2.1 Group Actions on Graphs . . . 11
2.1.1 Graph Isomorphisms and Automorphisms . . . 11
2.1.2 Acting on a Graph . . . 13
2.2 The Cayley Graph . . . 14
2.2.1 Definition and First Examples . . . 14
2.2.2 Acting on a Cayley Graph . . . 17
2.3 Tiling a Tree . . . 19
2.3.1 G-Tilings . . . 19
2.3.2 Proof of Theorem 2.17 . . . 21

Introduction

Free groups were first studied in [Fuc80] as an example of Fuchsian groups, which are discrete subgroups of $PSL_2(\mathbb{Z})$, but they did not become relevant on their own until the late 1800s. Any group may be written using its group presentation (for example, $D_3 = \langle r, s | r^3 = s^2 = 1, rs = sr^2 \rangle$). In [Dyc82], von Dyck, who was first to express groups using a presentation, showed that free groups have the simplest possible one (n generators with no relations) and that all finitely generated groups are the quotient of a free group. The algebraic properties of free groups were investigated by Nielsen, who gave them their name in [Nie24], while Dehn, Reidemeister, and Schreier also played important roles in developing the field throughout the first half of the 1900s.

The simple nature of free groups is equal parts useful and deceiving. Consider The Word Problem.

> **The Word Problem**: For a group G, let g be an element expressed as a product of G's generators. Find an algorithm that shows, in finitely many steps, whether or not g is the identity element.

While it is known that this problem is undecidable for arbitary groups, we will show in the next section that it is solvable if G is a free group. This does not mean that free groups are a trivial class of groups. In fact, understanding their algebraic properties has been an area of study for over a hundred years. To demonstrate this, we consider two questions.

First, suppose we are given the presentation of an arbitrary group G. Can we easily check whether or not G is a free group? This question was answered in the negative in [Ady55; Rab58], where it was shown that such a question is undecidable.

The second question is one that is often asked when a property of a structure is first defined: do the substructures of this structure also have this property? In particular, are the subgroups of a free group also free groups? This was answered in the positive in the 1920s, first in [Nie21] for finitely

generated subgroups, then in [Sch28] in the full form. However, such proofs were not trivial, and required high level topological or algebraic methods to get to a rigorous proof. Understanding the proofs would require understanding very high level results in group theory, far beyond the level of an undegraduate course in abstract algebra.

Answers to the both of these questions do exist in simpler forms within the field of Combinatorial/Geometric Group Theory, which in part seeks to understand a group's algebraic properties by looking at how it acts on a space. This extra assumption on how a group acts upon a space makes these results much easier to understand at the undergraduate level.

An answer to the first question can be found thanks to the Ping-Pong Lemma, which describes a sufficient condition for a group to be free, so long as it acts in a certain way on a set X. The result is often attributed to Felix Klein [Har00], who used it in the late 1800s, but is also called Schottky's Lemma, after Freidrich Schottky.

Theorem 0.1 (The Ping-Pong Lemma)**.** *Let $\{g_1, g_2, \ldots, g_n\}$ generate a group G, which acts on a set X. If*

(i) X contains n subsets $X_1, \ldots, X_n$ such that $X_i \cap X_j = \varnothing$, and

(ii) $g_i^k(X_j) \subset X_i$ for all nonzero powers k and $i \neq j$.

Then G is isomorphic to F_n.

More interestingly, an answer to question two is a corollary of a result proven by Serre in [Ser80], who showed that free groups may be classified by how they act on trees.

Theorem 0.2 (Serre's Theorem, 1980)**.** *G is a free group if and only if it acts freely on a tree without inversion.*

In this paper, we present full proofs and explanations of both results. First, we prove the Ping-Pong Lemma and use it to show that free groups of any rank, including one of infinite rank, lie within the matrix group $SL_2(\mathbb{Z})$. Afterwards, we prove both sides of Serre's Theorem. In proving the forwards direction, we define a special graph for our free group to act on called the Cayley graph, and then show that free groups acts freely on their corresponding Cayley graphs. To prove the reverse direction, we show that every group acting on a tree induces a special tiling of the tree, before finding a free generating set using this tiling.

§1 The Ping Pong Lemma

1.1 Definitions

We begin by rigorously defining what a free group is, beginning with an algebraic definition born from category theory, before transitioning to a combinatorial definition that will allow us to work with free groups more easily in later sections.

1.1.1 The Universal Property.

For us to work with free groups, we first need to understand what it means for a group to be "free."

Definition 1.1. Let F be a group. Then F is said to be **free** on a set S if the following holds: for any function f from S to a group G, there is a unique homomorphism $\varphi : F \to G$ such that the below diagram commutes (where ι denotes inclusion).

$$\begin{tikzcd} & F \arrow[d, dashed, "\varphi"] \\ S \arrow[ur, "\iota"] \arrow[r, "f"'] & G \end{tikzcd}$$

We call S the **generating set** of F.

The condition that F must satisfy is called the **universal property**, and plays an important role in developing free structures in category theory.

Example 1.2. Take $S = \{A, B\} \subset SL_2(\mathbb{Z})$, where

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

Then consider the group $F = \langle A, B \rangle$ generated by A and B. We will show later that F is free on S.

A priori, it is not obvious that for every set there exists a group that is free on it. However, not only is there such a group, but this group is unique up to isomorphism.

Before proving this, we introduce the concept of "reduced words." In the above example, elements of F are finite products of A, B, A^{-1}, and B^{-1}, such as $A, B^{-1}, ABAB, A^2B^3A$, etc. Such elements are called **words** in $\{A, B\}$. F does not contain an element like $A^{-1}A^3$, as this is just A^2. In general, there is not word in F containing a element right next to its inverse, and because of this, all elements of F are said to be **reduced words**. As another example, the word $ABB^{-1}A$ is not reduced, but A^2 is (we got A^2 by removing BB^{-1} from the original word, as this is just the identity matrix). We can combine words using concatenation. For example, concatenating AB and $B^{-1}A^2$ gives us $ABB^{-1}A^2$. This is not a reduced word, but it can be made reduced by removing the BB^{-1}, giving us A^3.

Theorem 1.3. *For any set S, there is a free group F_S that is free on S.*

F_S is also unique up to isomorphism; if F_S^1, F_S^2 are both free on S, with inclusion maps given by $\iota_1 : S \to F_S^1, \iota_2 : S \to F_S^2$, then there are unique isomorphisms $\hat{\iota_2} : F_S^1 \to F_S^2$ and $\hat{\iota_1} : F_S^2 \to F_S^1$ such that

$$\hat{\iota_2} \circ \iota_1 = \iota_2, \quad \hat{\iota_1} \circ \iota_2 = \iota_1, \quad \hat{\iota_1}^{-1} = \hat{\iota_2}$$

Proof. We let F_S be the set of reduced words in S, with the operation given by concatenation and reduction. Now, for any $f : S \to G$, where G is a group, we define φ as follows: we send the empty word $\varnothing$ to 1_G, the identity element of G, and for any $s \in S$,

$$\varphi(s) = f(s)$$

where we think of s as a word consisting of a single letter in F_S. φ can then be uniquely extended to all other words by the fact that it is a homomorphism.

We now show uniqueness. Setting $G = F_S^1$ and $f = \iota_1$ in the diagram from Definition 1.1 gives us

$$\begin{tikzcd} & F_S^1 \arrow[d, dashed, "\mathrm{id}_{F_S^1}"] \\ S \arrow[ur, "\iota_1"] \arrow[r, "\iota_1"'] & F_S^1 \end{tikzcd}$$

Similarly, setting $G = F_S^2$ and $f = \iota_2$, we get

$$\begin{tikzcd} & F_S^1 \arrow[d, dashed, "\hat{\iota_2}"] \\ S \arrow[ur, "\iota_1"] \arrow[r, "\iota_2"'] & F_S^2 \end{tikzcd}$$

from which we get that $\hat{\iota_2} \circ \iota_1 = \iota_2$. Swapping F_S^1 and F_S^2 and their respective maps gives us $\hat{\iota_1}$ such that $\hat{\iota_1} \circ \iota_2 = \iota_1$. Notice that $\hat{\iota_1} \circ \hat{\iota_2} : F_S^1 \to F_S^1$ satisfies

$$\hat{\iota_1} \circ \hat{\iota_2} \circ \iota_1 = \hat{\iota_1} \circ \iota_2 = \iota_1$$

satisfying the same equation that $\mathrm{id}_{F_S^1}$ does. By uniqueness, we get that $\hat{\iota_1} \circ \hat{\iota_2} = \mathrm{id}_{F_S^1}$. A similar argument shows that $\hat{\iota_2} \circ \hat{\iota_1} = \mathrm{id}_{F_S^2}$. This completes the proof. $\square$

The presence of a set S seems to complicate our definition, as we must define free groups in terms of this generating set. However, we can greatly reduce the number of sets we must consider by making a key observation. Suppose S, T are sets such that $|S| = |T|$, and let F_S, F_T be the corresponding free groups. Recall that we construct each free group by sending each element of the generating set to a letter in the free group, and define the homomorphisms φ by how they map these letters. As S, T are the same size, there is a bijective map $h : S \to T$. More importantly, this map lifts to one on the free groups

$$\psi : F_S \to F_T, \quad \psi(\iota_S(s)) = \iota_T(h(s))$$

As h is a bijection on the generating sets, and each free group is uniquely determined by the letters these sets map to, ψ is an isomorphism of the groups. We have just proven

Theorem 1.4. *If $|S| = |T|$, then $F_S \cong F_T$.*

Because of this, we only need to consider free groups generated by sets of the form $\{1, 2, \ldots, n\}$; the size of this set is called the free group's **rank**. We will thus denote these groups by F_n, called the **free group of rank** n. For the purposes of this paper, we will restrict our study to free groups of countable rank, meaning we consider F_n for all $n \in \mathbb{N}$, as well as F_∞, the free group with generators x_i for all natural i.

The construction of free groups makes them naturally important in the study of arbitrary groups. The first to notice this fact was Walther von Dyck, one of the founders of Combinatorial Group Theory, and the first to study groups via generators and relations.

Definition 1.5. Let S be a set of size n, and let $R \subseteq F_n$. Then we define

$$\langle S|R\rangle = F_n/N(R)$$

where $N(R)$ is the normal subgroup generated by R. We then say that a group G has a **presentation** given by $\langle S|R\rangle$ if $G \cong \langle S|R\rangle$. If so, we call S the set of **generators** of G, and R the set of **relations** on G.

It was through this study of groups through generators and relations that he found the usefulness of free groups:

Theorem 1.6 (von Dyck, 1882)**.** *Every finitely generated group G is the quotient of a free group.*

Proof. Let G have a presentation given by $\langle g_1, g_2, \ldots, g_n|R\rangle$. Let $S = \{1, 2, \ldots, n\}$ and let F_n be free on S. Consider the map

$$f : S \to G, \quad f(i) = g_i$$

Then by the universal property, let $\varphi : F_n \to G$ be the unique homomorphism associated with f. It follows that

$$\ker \varphi = \{g = g_1^{e_1} \cdots g_n^{e_n} : g = 1\} = R$$

In other words, $\ker G$ is the set of all relations on G. Thus, we get that

$$G \cong F_n/\ker \varphi. \qquad \square$$

Because of von Dyck's Theorem, we are able to define free groups in a very simple manner. By construction, the relations within a free group F_n are strictly those saying that $x_i x_i^{-1} = 1$ for any $i \in \{1, 2, \ldots, n\}$. Thus, free groups have the simplest possible presentation, and can be defined as just a set of generators with no relations:

Definition 1.7. The **free group of rank** n is the group F_n, the set of all reduced words in symbols $\{a_1, \ldots, a_n\}$, with operation given by concatenation and reduction. Its presentation is

$$\langle a_1, a_2, \ldots, a_n\rangle$$

and the identity is the empty word $\varnothing$.

We have now translated the algebraic definition of a group, that which uses the universal property, to one using reduced words, which are much easier to work with.

Using this definition, we can show that the Word Problem is solvable for free groups. A free group has no relations between the generators, apart from the obvious one that $a_i a_i^{-1} = 1$. Thus, given an arbitrary element of a free group, one can just check if any of these pairs exist, and eliminate them one at a time until none remain; if generators are still present, the element is not the identity, and if no generators are left, the element is the identity. Thus, while The Word Problem is generally undecidable, it is easily solvable in the case of free groups.

1.1.2 The Rank of Subgroups of F_2. For a group G, the **rank** of G, denoted $\text{rank}(G)$, is defined as

$$\min\{|S| : S \subseteq G, S \text{ generates } G\}$$

For free groups F_n, the rank is just the size of the set that F_n is free on, which we have already established is n.

Rank is a group-theoretic analogue of dimension for vector spaces. A well-known property of dimension is that for a vector space V, any subspace W will have at most dimension $\dim V$, which makes intuitive sense. However, for a group G and subgroup H of G, it is not necessarily the case that $\text{rank}(H) \leq \text{rank}(G)$.

This is especially true for the subgroups of free groups, which can sometimes have a much larger rank than the free group itself. In particular, F_2 not only contains subgroups of larger rank, but they contain subgroups of every possilbe rank larger than 2, including subgroups of countably infinite rank!

Theorem 1.8. *(a) For all $n > 2$, $F_n \subset F_2$.*

(b) $F_\infty \subset F_2$, where

$$F_\infty = \langle a_1, a_2, a_3, \ldots \rangle$$

is the free group of countably infinite rank.

Proof. (a) We write $F_2 = \langle a, b \rangle$, and define

$$\begin{aligned} w_1 &= a \\ w_2 &= b^{-1}ab \\ w_3 &= b^{-2}ab^{-2} \\ &\vdots \\ w_n &= b^{-n}ab^n \end{aligned}$$

With the operations of concatenation and reduction, these form the generating set of a group. It thus suffices to show that there are no relations between these generators. First notice that for any w_i and any integer k:

$$\begin{aligned} w_i^k &= (b^{-i}ab^i)(b^{-i}ab^i)\cdots(b^{-i}ab^i) \\ &= b^{-i}a(b^ib^{-i})a(b^ib^{-i})a\cdots a(b^ib^{-i})ab^i \\ &= b^{-i}a^kb^i \end{aligned}$$

Now, let $w = w_{i_i}^{k_1} \cdots w_{i_l}^{e_l}$ be an arbitrary word. We have that

$$\begin{aligned} w &= w_{i_i}^{k_1} \cdots w_{i_l}^{e_l} \\ &= (b^{-i_1}a^{e_1}b^{i_1})(b^{-i_2}a^{e_2}b^{i_2})\cdots(b^{-i_l}a^{e_l}b^{i_l}) \\ &= b^{-i_1}a^{e_1}b^{i_i-i_2}a^{e_2}b^{i_2-i_3}\cdots b^{i_{l-1}-i_l}a^{e_l}b^{i_l} \end{aligned}$$

and this is fully reduced. Moreover, it clearly cannot be expressed as the reduced product of any other set other product of the generators. Therefore, we conclude that there are no relations between the w_i's, and hence they generate the free group F_n, so $F_n \subset F_2$.

(b) The proof is identical to (a), with the generating set of F_∞ being the set of w_i for all $i \in \mathbb{N}$.

□

1.2 The Ping-Pong Lemma

As stated previously, given a group presentation $G = \langle S|R\rangle$, there is no algorithm running in finite time that can determine if this group is free. However, if we provide additional information about how G acts on a set X, we can determine if this group is free. This result is known as the Ping-Pong Lemma, and is one of the foundational results in Combinatorial Group Theory. It was famously used by Jacques Tits in [Tit72] to prove the result now knows as the Tits Alternative.

Before introducing the result, we define what it means for a group to act on a set:

Definition 1.9. Let G be a group and X a set. A **group action** of G on X is a map $G \times X \to X$ defined as

$$(g, x) \mapsto gx$$

such that

1. $1x = x$ for all $x \in X$.
2. $g(hx) = (gh)x$ for all $g, h \in G, x \in X$.

If such a map exists, we say that G **acts on** X.

1.2.1 Statements and Proofs. There are many different statements of the Ping-Pong Lemma for free groups, free products, etc. We focus on a relatively simple version found in [Mar17]. We start with a special case:

Theorem 1.10 (The Ping-Pong Lemma for Two Players)**.** *Suppose a, b generate a group G, and suppose that G acts on a set X. If*

1. *X has disjoint nonempty subsets X_a, X_b; and*
2. *for all powers $k \neq 0$, $a^k(X_b) \subset X_a$ and $b^k(X_a) \subset X_b$.*

Then G is isomorphic to a free group of rank 2.

Proof. As G has only 2 generators, it suffices to show there are no relations in G. Let

$$g = a^{e_1} b^{e_2} \cdots b^{e_{k-1}} a^{e_k}$$

be a word in G. We show that $g \neq 1$ by showing that $gp \neq p$ for some $p \in X$. Let $p \in X_b$ and consider gp. Notice that a^{e_k} will send p to X_a, then $b^{e_{k-1}}$ will send $a^{e_k}p$ back to X_b. We go back and forth from X_a, to X_b, etc.

At the end, a^{e_1} takes $b^{e_2} \cdots b^{e_{k-1}} a^{e_k} p$ to X_a. As $X_a \cap X_b = \varnothing$, and $gp \in X_a$ while $p \in X_b$, $gp \neq p$, meaning $g \neq 1$.

We conclude by showing that any nontrivial word in G is conjugate to a word of the form $a^{e_1} b^{e_2} \cdots b^{e_{k-1}} a^{e_k}$. That is, for every nontrivial $h \in G$, there is a $y \in G$ such that

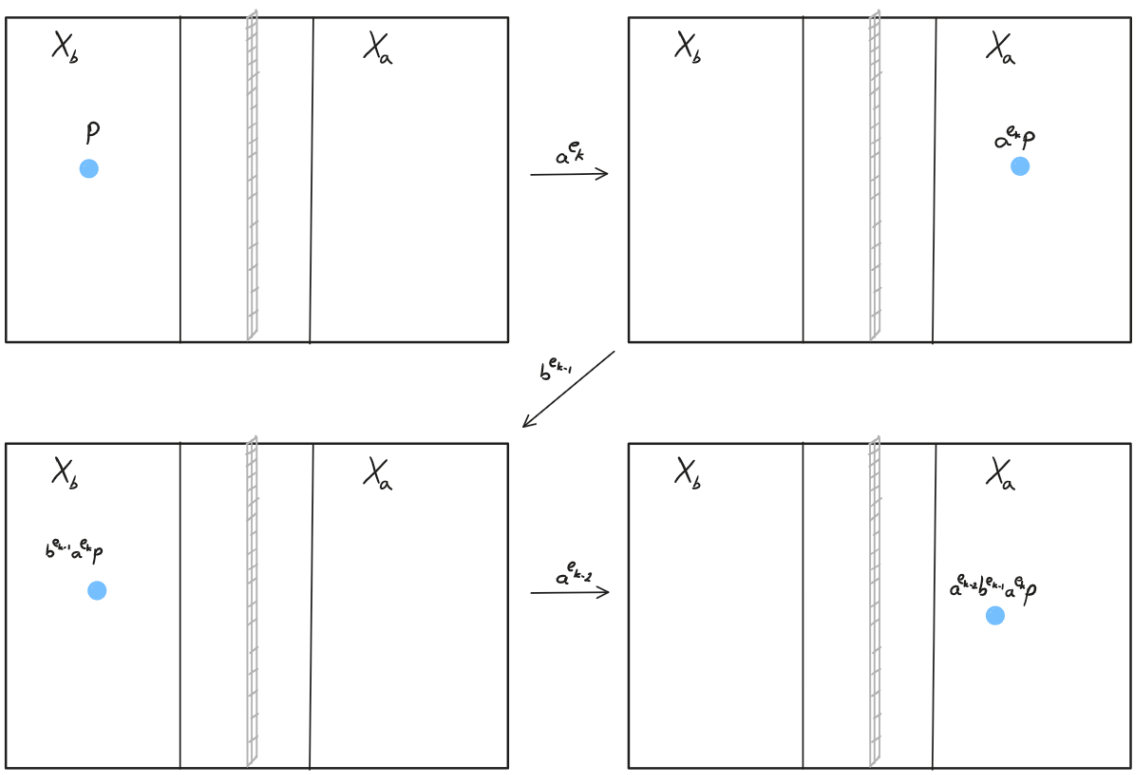
$$yhy^{-1} = a^{e_1} b^{e_2} \cdots b^{e_{k-1}} a^{e_k}$$

Let h be a nontrivial reduced word in $\{a, b, a^{-1}, b^{-1}\}$. For the purpose of brevity, we will let $*$ denote a nonzero exponent. We have several cases:

1. If $h = a^*b^* \cdots b^*a^*$, take $y = 1$.
2. If $h = b^*a^* \cdots b^*a^i$ or $h = a^ib^* \cdots b^*a^*$, take $y = a^j$ for some $j \neq i$.
3. If $h = b^*a^* \cdots a^*b^*$, take $y = a$.

This tells us that any word in G is conjugate to a nontrival element of G. As all conjugates of 1 are trivial, this means that every word in G must be nontrivial. Thus, there are no relations between elements of G, so G is isomorphic to a free group of rank 2. $\square$

The section of the proof where we bounce between X_a and X_b should give a good idea as to why this is called the Ping-Pong Lemma. p starts in X_b, then is sent to X_a, then back to X_b, then back to X_a, etc. It's like watching a ping-pong game, where the ball is p, X_a and X_b are the sides of the table, and the ball is "hit around" by g.



We can extend this proof for rank 2 to find free groups of any finite rank:

Theorem 1.11 (The Ping-Pong Lemma for n Players)**.** *Suppose $\{g_1, \ldots, g_n\}$ generate a group G, and $G \curvearrowright X$. If*

1. *X has pairwise disjoint nonempty subsets $X_1, \ldots, X_n$, and*
2. *for all powers $k \neq 0$, $g_i^k(X_j) \subset X_i$ for all $i \neq j$.*

Then G is isomorphic to a free group of rank n.

Proof. Let $g = g_{i_1}^{e_1} g_{i_2}^{e_2} \cdots g_{i_k}^{e_k}$ be a nontrivial reduced word in G, and let $p \in X_j$ where $j \neq i_k$. It follows that $gp \in X_i$, $i \neq j$, and thus $g \neq 1$. This suffices as all elements of G are conjugate to an element of the above form. $\square$

After seeing this, one may ask if this forms both a necessary and sufficient condition for a group to be free. That is, given a group G acting on a set X, is it free if and only if the properties of X and the group action described in the Ping-Pong Lemma apply? The answer is yes, because for any free group, there is always a set and group action on that set fulfilling these conditions.

Theorem 1.12. *There is a set X with subsets X_i and a group action of F_n on X fulfilling the conditions of the Ping-Pong Lemma.*

Proof. We let

$$X = \{1, 2, \ldots, n\}$$

and define $X_i = \{i\}$ for all i. This gives us n pairwise disjoint nonempty subsets, fulfilling the first condition of the Ping-Pong Lemma.

We define our group action of F_n on X with respect to generators. Suppose F_n is generated by $a_1, a_2, \ldots, a_n$. Then for any $x \in X$, we define

$$a_i x := i$$

Clearly, if $k \neq 0$ and $i \neq j$, then

$$a_i^k j = i$$

so $a_i^k(X_j) \subset X_i$, fulfilling the second condition of the Ping-Pong Lemma. □

1.2.2 An Application: $SL_2(\mathbb{Z})$. The Ping-Pong Lemma is a powerful tool for finding free subgroups of certain groups. In finding them, we can find a large class of free subgroups thanks to the Schreier Index Formula.

Recall that $SL_2(\mathbb{Z})$ is the group of 2×2 matrices of determinant 1. Let $G = \langle A, B \rangle$ be a subgroup of $SL_2(\mathbb{Z})$ generated by

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

Lemma 1.13. *For any power $k \neq 0$,*

$$A^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}, \quad B^k = \begin{pmatrix} 1 & 0 \\ 2k & 1 \end{pmatrix}$$

Proof. First assume $k > 0$. We proceed by induction. The base case $k = 1$ is obvious. Now assume that the claim holds on k and consider $k + 1$. Then we have that

$$\begin{aligned} A^{k+1} &= (A^k)A \\ &= \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 + 2k \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2(k+1) \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Similarly,

$$\begin{aligned} B^{k+1} &= (B^k)B \\ &= \begin{pmatrix} 1 & 0 \\ 2k & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 2k + 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 2(k+1) & 1 \end{pmatrix} \end{aligned}$$

as desired. For negative exponents, we first note that

$$\begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2k \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 2k & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2k & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and this completes the proof since $(A^k)^{-1} = A^{-k}$. $\square$

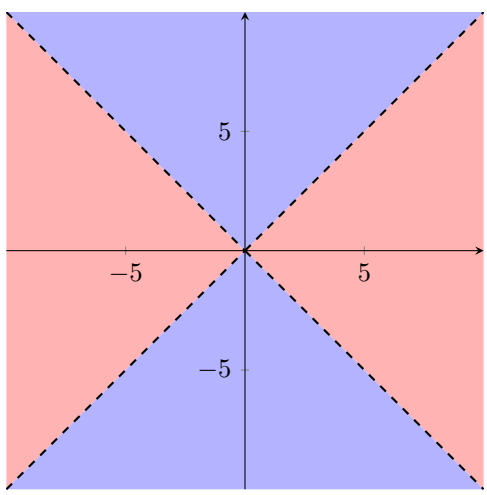
We claim that G is isomorphic to a free group of rank 2. To do this, we define the action of G on $\mathbb{Z}^2$ by standard matrix multiplication:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

and we define our sets as

$$X_A = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : |x| > |y| \right\}, \quad X_B = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : |x| < |y| \right\}$$

As a visual, X_A is shown below in red, while X_B is shown in blue.



We claim that X_A, X_B meet the criteria of the Ping-Pong Lemma. First note that these sets are clearly disjoint. Next, let $\vec{x} \in X_a$, so $|x| < |y|$. Using Lemma 1.13, we have that for any power $k \neq 0$,

$$A^k \vec{x} = \begin{pmatrix} x + 2ky \\ y \end{pmatrix}$$

and by the Reverse Triangle Inequality,

$$\begin{aligned} |x + 2ky| &\geq ||x| - |2ky|| \\ &\geq |x| - |2ky| \\ &= |x| - |2k||y| \\ &> |2k||y| - |x| \\ &\geq 2|y| - |x| \\ &= |y| + (|y| - |x|) \\ &> |y| && (|y| - |x| > 0) \end{aligned}$$

so $A^k\vec{x} \in X_B$, meaning $A^k(X_A) \subset X_B$. Similarly, if $\vec{x} \in X_B$, then for any power $k \neq 0$,

$$B^k\vec{x} = \begin{pmatrix} x \\ 2kx + y \end{pmatrix}$$

and we get that

$$\begin{aligned} |2kx + y| &\geq ||2kx| - |y|| \\ &\geq |2kx| - |y| \\ &= |2k||x| - |y| \\ &\geq 2|x| - |y| \\ &= |x| + (|x| - |y|) \\ &> |x| \qquad (|x| - |y| > 0) \end{aligned}$$

so $B^k\vec{x} \in X_A$, meaning $B^k(X_B) \subset X_A$. The Ping-Pong Lemma then applies, so G is isomorphic to a free group of rank 2. In addition to showing that F_2 is a subgroup of $SL_2(\mathbb{Z})$, by Theorem 1.8, this also tells us that $F_n \subset SL_2(\mathbb{Z})$ for any $n > 2$.

We can generalize this result slightly to find other free subgroups of $SL_2(\mathbb{Z})$. Set

$$A = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ m & 1 \end{pmatrix}$$

where $m > 2$. Using the exact same methods as before, we may show that $G = \langle A, B \rangle \subset SL_2(\mathbb{Z})$ is isomorphic to a free group of rank 2.

§2 Serre's Theorem

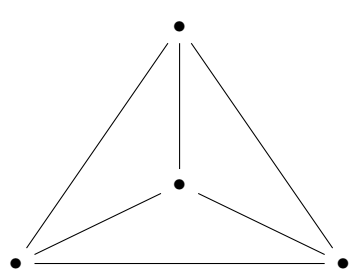
2.1 Group Actions on Graphs

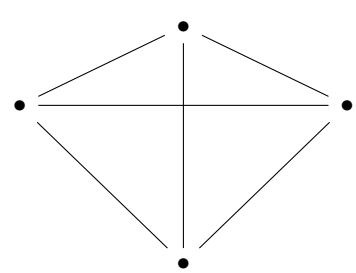
To begin understanding Serre's Theorem, we first need to define what it means for a group to act on a graph, as well as when such an action is free.

For our purposes, a **graph** Γ is a combinatorial object given by a tuple (V, E), where V is a set of **vertices**, and $E \subseteq V \times V$ is a set of **edges**. We will sometimes denote the vertex and edge sets of Γ by $V(\Gamma)$ and $E(\Gamma)$ respectively.

A graph can be undirected (meaning if $(v, w) \in E$, then $(w, v) \in E$), directed (meaning at most one of (v, w) or (w, v) is in E), can contain loops (edges of the form (v, v)), and can have multiple edges going between two vertices. We may also have labels or colours on certain edges and vertices in a graph to distinguish them; this will come in handy later when defining the Cayely Graph of a group.

2.1.1 Graph Isomorphisms and Automorphisms. For many, there is an intuitive definition of when two graphs are isomorphic: they look the same. For example, these two isomorphic graphs represent different perspectives of a tetrahedron:





Definition 2.1. We say that two graphs $\Gamma_1 = (V_1, E_1), \Gamma_2 = (V_2, E_2)$ are **isomorphic** if and only if there is a bijection $f : V_1 \to V_2$ that preserves edge adjacencies. In other words,

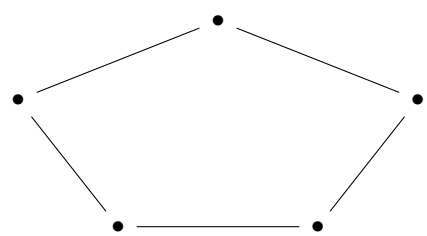
$$(v, w) \in E_1 \iff (f(v), f(w)) \in E_2$$

Such an f is called a **graph isomorphism**.

If we have additional structure on our graphs, such as labellings, colours, or directed edges, then the isomorphism must preserve these structures. Blue edges must go to blue edges, directed edges to directed edges, etc.

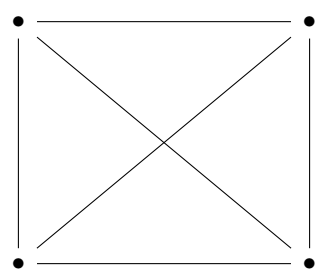
We can also define an isomorphism from a graph to itself. Such isomorphisms are called **automorphisms**, and will play an important role in proving one direction of Serre's Theorem. Automorphisms are the maps which represent the symmetries of a graph. The set of symmetries of a graph Γ forms a group with function compostition, which is denoted $Aut(\Gamma)$.

Example 2.2. Consider the graph representing a regular pentagon:



There are several obvious automorphisms, such as rotation by $2\pi/5$, as well as reflections through the lines passing through both the center of the pentagon and any vertex. All in all, we see that the automorphism group of this graph is D_5, the dihedral group of a pentagon. If this graph were directed, with edges going clockwise around the pentagon, then we would not be able to make reflections, and thus the automorphism group would be $\mathbb{Z}_5$.

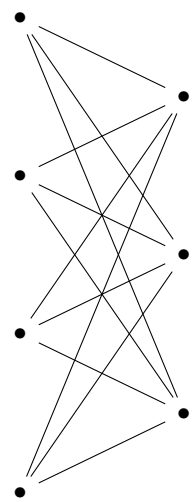
Example 2.3. Consider the graph K_4, the complete graph on 4 vertices:



Notice how every vertex is connected to every other vertex. Thus, swapping any two vertices preserves the graphs structure and is an automorphism. More generally, any permutation of these 4 vertices will count as an automorphism. Thus, the automorphism group of K_4 is S_4, the symmetric group on 4 elements. More generally,

$$Aut(K_n) = S_n$$

Example 2.4. Consider the complete bipartite graph $K_{4,3}$:



Like K_4, we can permute the vertices on the left side and not affect the graph's structure, since they are all connected to every vertex on the right side. Similarly, we can permute the vertices on the right side as well. However, we cannot swap a vertex on the left with one on the right, as they have different degrees. Thus, we see that $Aut(K_{4,3}) = S_4 \times S_3$. More generally, if $n \neq m$,

$$Aut(K_{n,m}) = S_n \times S_m$$

If we had $K_{n,n}$, then we would have the addition symmetry of swapping the left and right sides (we cannot swap individual vertices since that would change the edge adjacencies). Thus,

$$Aut(K_{n,n}) = (S_n \times S_n) \rtimes \mathbb{Z}_2$$

2.1.2 Acting on a Graph. Now having defined what graphs are, as well as their symmetries, we are ready to define what it means for a group to act on a graph.

Definition 2.5. Let G be a group and Γ a graph. A homomorphism $G \to Aut(\Gamma)$ is called an **action** of G on Γ. If such a homomorphism exists, then we say that G **acts on** Γ.

Using this definition, we can restate our previous examples as group actions:

1. D_n acts on the graphs of regular n-gons by rotation and reflection. If the graph is directed, $\mathbb{Z}_n$ acts on it by rotation only.

2. S_n acts on K_n by permuting the vertices.

3. $S_n \times S_m$ acts on $K_{n,m}$ by permuting the left and right vertex sets, while $(S_n \times S_n) \rtimes \mathbb{Z}_2$ acts on $K_{n,n}$ by permuting the vertex sets and swapping them.

We conclude this section by defining what it means for such an action to be free, while also introducing an important notion that will become more relevant later.

Definition 2.6. Let G act on a graph Γ. We say that the group action is **free** if for any vertex v, $g \cdot v = v$ if and only if $g = 1$. We say that G acts **without inversion** if for any edge $(v, w) \in E$ and $g \in G$, either $g \cdot v \neq w$ or $g \cdot w \neq v$.

Example 2.7. S_4 acts freely on K_4, but it does not act without inversion. On the other hand, $\mathbb{Z}_6$ acts without inversion on the graph of an equilateral triangle, but not freely; the element 3 corresponds to rotating by $2\pi/3$ three times, which is the same as doing nothing.

The idea of acting without inversion simply says that G does not flip any edge of Γ. For any group G acting on a graph Γ, the action can always be made into one without inversion by acting on Γ's **barycentric subdivision**, Γ', created by introducing a vertex to every edge in Γ. An example is shown below:

2.2 The Cayley Graph

While all of these are cool and interesting examples of group actions, as they demonstrate the mathematical definition of graph symmetry, we have yet to see free groups in this context. Is there a graph for which the automorphism group of that is graph is, say, F_2? What graphs do free groups act on? Do they act freely?

These questions are answered by the Cayley graph. It was first introduced in the context of finite groups by Cayley in [Cay78].

2.2.1 Definition and First Examples.

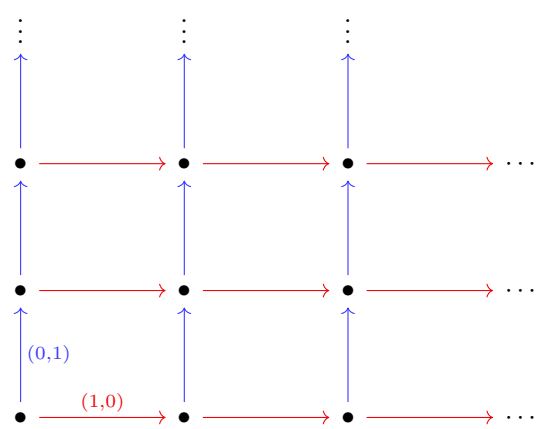
Definition 2.8. Let G be a graph generated by a set S. The **Cayley graph of G with respect to S** is the graph $\Gamma(G, S)$, defined as follows: The vertex set is G. Each $s \in S$ is assigned a colour c_s. Then for each vertex g, there is a c_s-coloured edge from g to sg.

One can think of the Cayley graph of a group as a visual representation of that group.

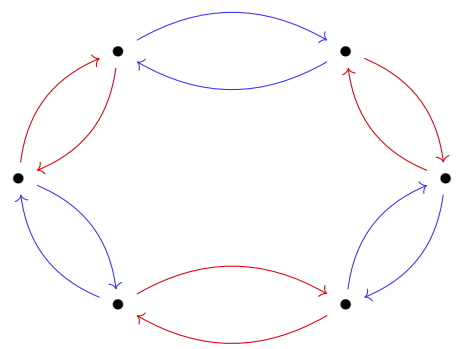
Example 2.9. Consider $G = \mathbb{Z} = \langle 1 \rangle$. The Cayley graph can be thought of as a line of points, each representing an integer, with a single $+1$ edge entering and exiting each.

$$\cdots \longrightarrow \bullet \xrightarrow{1} \bullet \longrightarrow \bullet \longrightarrow \cdots$$

Similarly, for $G = \mathbb{Z}^2 = \langle (0,1), (1,0) \rangle$, the Cayley graph is an integer lattice, with red edges going horizontally representing $(0,1)$, and blue edges going vertically representing $(1,0)$.



Example 2.10. Consider $S_3 = \langle (1\ 2), (2\ 3) \rangle$. It's Cayley graph is shown below, with red edges representing $(1\ 2)$ and blue edges representing $(2\ 3)$.



Example 2.11. Here's a fun example of a non-planar Cayley graph. The **Heisenberg Group** H is the group of 3×3 upper triangular matrices of the form

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

where $a, b, c \in \mathbb{R}$, under the operation of matrix multiplication. This group arises in the study of certain quantum systems. If we restrict a, b, c to the integers, then we get the **Discrte Heisenberg Group**, $H_3(\mathbb{Z})$, which is generated by matrices

$$x = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

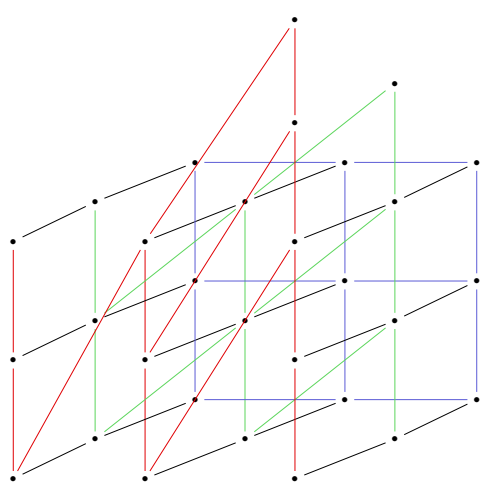
and has relations

$$z = xyx^{-1}y^{-1}, \quad xz = zx, \quad yz = zy$$

where

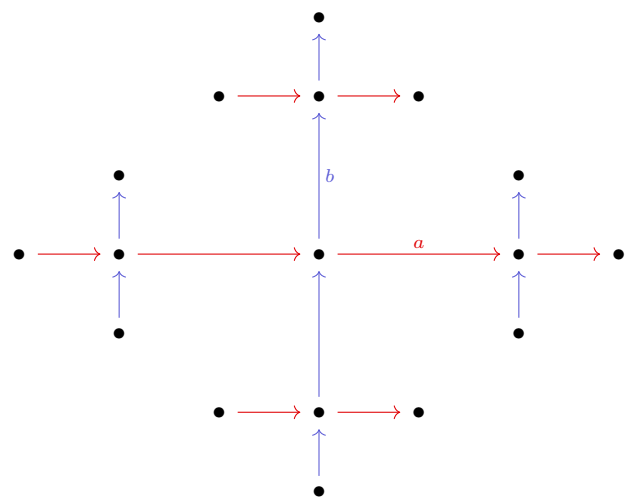
$$z = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A portion of $H_3(\mathbb{Z})$'s Cayley graph is shown below. Notice how it is shown embedded into 3D space. Edges going vertical represent z, those going horizontally represent y, and those going "out of the page" represent x; the colours are a visual aid due to the cluttered nature of the graph.

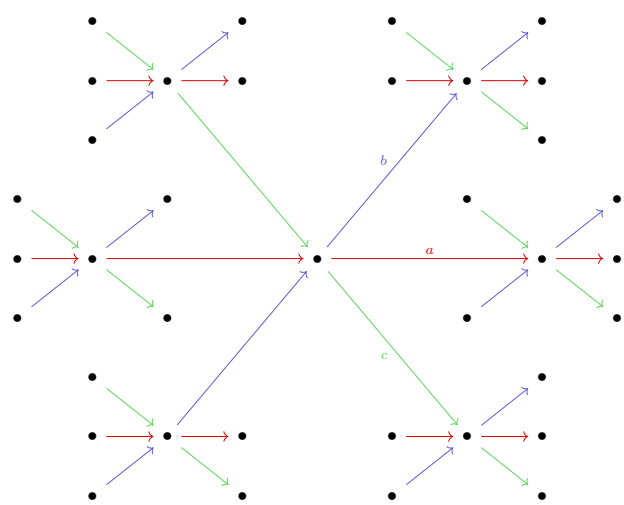


The Cayley graph of F_n will be an infinite graph, as F_n is itself infinite, but will contain a very interesting structure that, for our purposes, will become very important.

For simplicity's sake, we will just show the Cayley graphs of F_n for small values of n. $F_1 = \mathbb{Z}$, the Cayley graph of which we've already seen in the previous subsection. For $F_2 = \langle a, b \rangle$, we get the following graph, which has a very nice symmetrical structure akin to a cross:



For $F_3 = \langle a, b, c \rangle$, we get hexagonal symmetry.



Remark: Some readers may be somewhat uneasy about the definition for Cayley graph given above because it is in terms of a given generating set. A group can have multiple generating sets. For instance, $\mathbb{Z} = \langle 1 \rangle = \langle 2, 3 \rangle$. As such, one should be skeptical of whether or not our definition is well-defined. Of course, it is, but to show that we need to do a little bit of work. Consider $\mathbb{Z}^2$. We know that this acts on its Cayley graph by translation horizontally or vertically, but it also acts similarly on $\mathbb{R}^2$, the Euclidean plane. Clearly theses spaces are not homeomorphic. However, if you look at these spaces from a far enough distance away, they do look very similar. Such a property has a special name: we say that the spaces are **quasi-isometric**. We will omit the proof of this. de la Harpe's book [Har00] is a great reference for this topic.

Another thing we mention but not prove is that a classification of Cayley graphs has been found:

Theorem 2.12 (Sabidussi's Theorem)**.** *An unlabeled, uncoloured, directed graph Γ is the Cayley graph of a group G if and only if it admits a simply transitive action of G by graph automorphisms.*

The proof can be found in [Sab58].

2.2.2 Acting on a Cayley Graph. One of the ways that G can act on itself is by left multiplication:

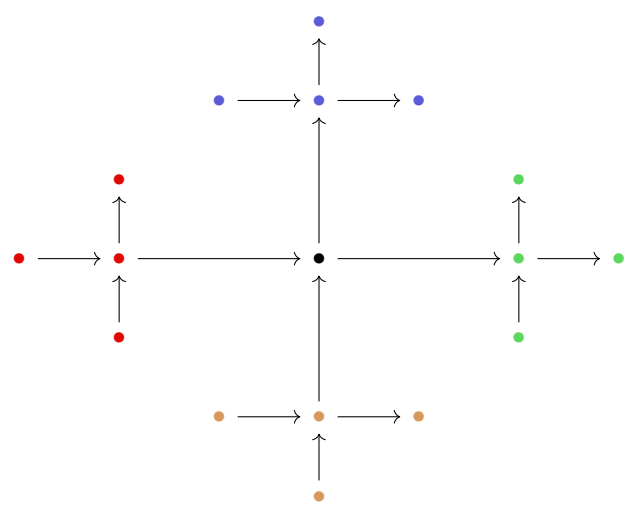
$$g \cdot h = gh$$

This can be reformulated to create an action on our Cayley graph. In particular, for each $g \in G$, we define the automorphism Φ_g of the Cayley graph by sending vertex h to vertex gh. This is indeed an automorphism as it preserves edge adjacency. If v, vh are connected by the edge h, then gv is connected to gvh by the edge h as well.

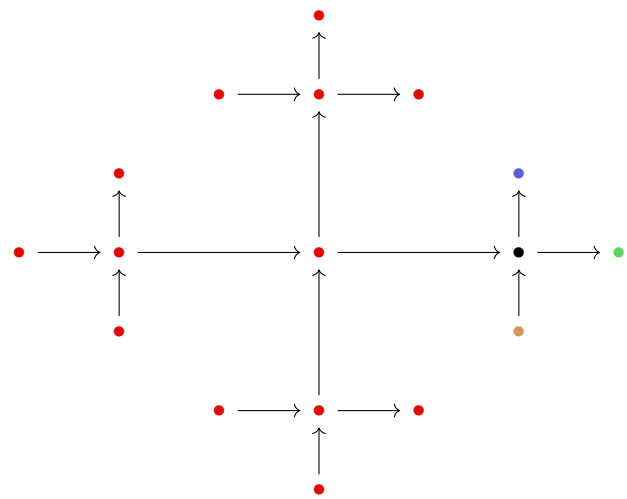
For non-free groups, this action can look like translation (as is evident in the case of $\mathbb{Z}$ and $\mathbb{Z}^2$), or rotation (as seen in the case for S_3). For free groups, we get a much more interesting set of symmetries. Consider F_2 acting on $\Gamma(F_2, \{a, b\})$ as above. Let's see what happens when we apply a to each vertex. Let X_a be the set of all words in F_2 that start with an a, and let $X_b, X_{a^{-1}}$, and $X_{b^{-1}}$ be defined similarly. Then the vertex set of $\Gamma(F_2, \{a, b\})$ may be partitioned into 5 groups: $\{1\}, X_a, X_b, X_{a^{-1}}$, and $X_{b^{-1}}$.

Multiplying on the left by a, we see that $a\{1\} = \{a\} \subset X_a$. Similarly, $aX_a, aX_b, aX_{b^{-1}} \subset X_a$. For any word starting with a^{-1}, multiplying on the left by a means we cancel out the starting a^{-1}. The new reduced word could either be 1 (if the original word was a^{-1}), or must start with either an a^{-1}, b, or b^{-1} (it cannot start with an a as it would mean the original word was not reduced). Thus, $aX_{a^{-1}} \subset \{1\} \cup X_{a^{-1}} \cup X_b \cup X_{b^{-1}}$.

If we draw this out, we get that these vertices:



get sent to these vertices:



This action of F_n onto its Cayley graph satisfies some key properties, which are proven in the lemmas below:

Lemma 2.13. *If $G = \langle S|R\rangle$, then there is a one-to-one correspondence between relations $r \in R$ and cycles in $\Gamma(G, S)$.*

Proof. First let $r \in R$ be some relation of the form

$$g_{i_1} g_{i_2} \cdots g_{i_k} = g_{j_1} g_{j_2} \cdots g_{j_l}$$

for some $g_{i_n}, g_{i_m} \in G$. We note that there is an edge path from 1 to $g_{i_1} g_{i_2} \cdots g_{i_k}$ by following the edges $g_{i_1}, g_{i_2}, \ldots, g_{i_k}$. Similarly, there is an edge path from 1 to $g_{j_1} g_{j_2} \cdots g_{j_l}$. As $g_{i_1} g_{i_2} \cdots g_{i_k} = g_{j_1} g_{j_2} \cdots g_{j_l}$, these paths end at the same vertex. Thus, taking the path from 1 to $g_{i_1}, g_{i_2}, \ldots, g_{i_k}$, then going in reverse along the path from 1 to $g_{j_1} g_{j_2} \cdots g_{j_l}$ produces a cycle in the Cayley graph.

Now, consider a cycle in $\Gamma(G, S)$ which hits vertices

$$1, g_1, g_2, g_3, \ldots, g_{n-1}, g_n, 1$$

Take some vertex g_i in this edge path. We thus get two edge paths from 1 to g_i, of the form $1, g_1, g_2, \ldots, g_{i-1}, g_i$ and $g_i, g_{i+1}, \ldots, g_n, 1$ repsectively. These paths end at the same vertex, g_i, and so we conclude that

$$g_1 g_2 \cdots g_{i-1} = g_n g_{n-1} \cdots g_{i+1}$$

which forms a relation in R, as required. □

Corollary 2.14. *$\Gamma(F_n, \{x_1, \ldots, x_n\})$ is a tree.*

Proof. As there are no relations in F_n, there can not be any cycles in the Cayley graph. □

Lemma 2.15. *G acts freely on its Cayley graph $\Gamma(G, S)$.*

Proof. We know G acts on its Cayley graph by left multiplication of vertices. Now, let $g \in G$ be such that $g \cdot v = v$ for all vertices v in the Cayley graph. Then taking $v = 1$, we get that

$$g \cdot 1 = 1 \implies g = 1$$

Conversely, if $g = 1$, then $g \cdot v = v$ for all vertices v. The action is thus free, as desired. □

Combining the above lemmas, we attain one of the directions for Serre's Theorem:

Theorem 2.16 (Serre's Theorem: First Direction)**.** *If G is free, then there is a tree T such that G acts on T freely and without inversion.*

Proof. Lemma 2.15 tells us that G acts freely on $\Gamma(G, S)$, which is a tree by Corollary 2.14. If G acts with inversion, then we can always pass the action on $\Gamma(G, S)$ to one on the barycentric subdivision $\Gamma(G, S)'$, which is also a tree. As G acts freely on its Cayley graph, it acts freely on that graph's barycentric subdivision, and acts on it without inversion, as desired. □

2.3 Tiling a Tree

We use this section to complete the proof of Serre's Theorem by proving

Theorem 2.17 (Serre's Theorem: Second Direction)**.** *If G is a group such that G acts on a tree T freely and without inversion, then G is a free group.*

We use a proof shown in [Mar17] which is similar to the one used in [Ser80]. Let T be a tree and let G be a group such that G acts freely and without inversion on a tree T. We will construct a free generating set for G by creating a special set of subtree of T's barycentric subdivision that are all translations of a "base" subtree. From here, we can create our free generating set from the subtrees that intersect with the base subtree.

2.3.1 G-Tilings.

Definition 2.18. A **tile** of a tree T is any subtree of T.

A **tiling** of T is a collection of tiles of T such that

1. The union of all tiles is T; and
2. two tiles intersect, at most, at a single vertex.

If G acts on T, and if

3. there is a tile T_0 such that the set of all tiles is $\{gT_0 : g \in G\}$

then we call the tiling a **G-tiling**.

It is not obvious a priori that T will always have a G-tiling. However, one can show that it always does.

Theorem 2.19. *Let G be a group and T a tree. If G acts on T freely, then T has a G-tiling.*

Note that even if G acted on T with inversion, we can always pass to the barycentric subdivision T' and get an action without inversion. For this reason, whenever we say T from here on out, we are referring to its barycentric subdivision. The reason for this will become relevant later in Lemma 2.21.

Now, how exactly will we create our G-tiling? We can do this using our group action. Let $v_0 \in V(T)$ be a vertex in the tree, and consider its orbit under the group action.

$$\mathcal{O}(v_0) = \{gv_0 : g \in G\}$$

For every vertex $v \in V(T)$, there is a $gv_0 \in \mathcal{O}(v_0)$ that is closest to v under the path metric. From this, we define a subgraph T_g as follows:

$$V(T_g) = \{v \in T : d(v, gv_0) \leq d(v, g'v_0)\ \forall g' \neq g\}$$

$$E(T_g) = \{(u, v) \in E(T) : u, v \in V(T_g)\}$$

In other words, T_g is a subgraph whose vertex set is all those vertices that are closest to gv in T under the path metric, and whose edge set is all those edges whose endpoints are both vertices in T_g.

Lemma 2.20. *Let G act on T. Then T_g is a tile for all $g \in G$.*

Proof. It suffices to prove that T_g is connected, as every connected subgraph of a tree is a subtree. Let $w \in V(T_g)$. There is a unique edge path w to gv; suppose it is of length n.

$$w_0 = w, w_1, w_2, \ldots, w_{k-1}, w_n = gv$$

Then $d(w_1, gv_0) = n - 1$. We claim that $w_i \in V(T_g)$ for all i. Starting with w_1, suppose by way of contradiction that $w_1 \notin V(T_g)$. Then there is a g' such that

$$d(w_1, g'v_0) = m < n - 1$$

But then going back on edge to w, we get that

$$d(w, g'v_0) \leq m + 1 < n$$

which contradicts the fact that $w \in V(T_g)$. Thus, $w_1 \in V(T_g)$. This logic can be repeated for all other vertices in the path, ensuring that all are in $V(T_g)$, as desired. This means that T_g is connected, hence a subtree of T, and thus, is a tile. □

Now that we know the T_g's are in fact tiles, we should also check that they cover all of T.

Lemma 2.21. *The union of all tiles T_g for all g is equal to T.*

Proof. It is evident that $\bigcup_{g \in G} V(T_g) = V(T)$, since every vertex must be closest to some gv_0. It is also evident that

$$\bigcup_{g \in G} E(T_g) \subseteq E(T)$$

So it suffices to show the reverse inclusion: every edge in T belongs to some T_g. Let $(u, w) \in E(T)$. Then by construction of T, the distance to $\mathcal{O}(v_0)$ from one of u, w is odd, while the distance from the other is even. Without loss of generality, suppose that

1. $d(u, \mathcal{O}(v_0))$ is even,
2. $d(w, \mathcal{O}(v_0))$ is odd,
3. $d(u, \mathcal{O}(v_0)) < d(w, \mathcal{O}(v_0))$; and
4. $u \in V(T_g)$.

By the Triangle Inequality, we have that

$$d(w, gv_0) \leq d(w, u) + d(u, gv_0) = d(u, gv_0) + 1$$

As the distance from w to $\mathcal{O}(v_0)$ is an integer, and cannot be smaller than or equal to $d(u, gv_0)$, the only possibility is that it is equal to $d(w, gv_0)$. Thus,

$$d(w, gv_0) \leq d(w, g'v_0) \ \forall g' \neq g$$

so $w \in V(T_g)$. We conclude that $(u, w) \in E(T_g)$, as desired. □

The second condition, that two tiles share at most a vertex, is somewhat obvious. Indeed, if (u, w) is in both $E(T_g)$ and $E(T_{g'})$, then u, w are of equal distance to gv_0 and $g'v_0$. But assuming u is of equal distance to gv_0 and $g'v_0$, w must be closer to one of gv_0 or $g'v_0$, since it is along the unique edge path from u to one of them.

The final condition can be proven using another lemma:

Lemma 2.22. *For $g, h \in G$, $gT_h = T_{gh}$.*

Proof. It suffices to show that g maps vertices in T_h to those in T_{gh}, as they're completely determined by vertices. Let $u \in V(T_h)$. Then for all $k \in G$,

$$d(u, hv_0) \leq d(u, kv_0)$$

Left multiplication by g^{-1} is a bijection of G, so we may write

$$d(u, hv_0) \leq d(u, (g^{-1}k)v_0)$$

For all $k \in G$. Now, as G acts on T, and T is a metric space using the path metric, the action of G preserves distances. Thus,

$$d(gu, (gh)v) \leq d(gu, kv)$$

so $gu \in T_{gh}$, as desired. □

Proof of Theorem 2.19. By Lemmas 2.20 and 2.21, the collection $\{T_g : g \in G\}$ is a tiling of T. Setting $h = 1$ and applying Lemma 2.22, all tiles are of the form gT_1 for some $g \in G$. □

We have thus shown that any group acting on a tree induces a G-tiling on the tree.

2.3.2 Proof of Theorem 2.17. Now that we know G induces a G-tiling on our tree T, we can use it to create a free generating set.

We let $\{T_g\}$ be the G-tiling found in Theorem 2.19. As G acts freely on T, $\mathrm{Stab}_G(v_0) = \{1\}$, so by the Orbit-Stabilizer Theorem,

$$|\mathcal{O}(v_0)| = 1 \cdot |G| = |G|$$

Thus the elements of G are in bijective correspondence with the elements of the orbit of v_0. In other words, every $g \in G$ corresponds to a tile gT_1. define

$$S = \{g \in G : (gT_1) \cap T_1 \neq \varnothing\}$$

to be the set of tiles that intersect with T_1.

Lemma 2.23. *S generates G*

Proof. First we show that if $s \in S$, so too is s^{-1}. Indeed let $s \in S$. Then there is a vertex w such that

$$\begin{aligned} &(sT_1) \cap T_0 = \{w\} \\ \implies &s^{-1}(sT_1) \cap (s^{-1}T_1) = \{s^{-1}w\} \\ \implies &T_1 \cap (s^{-1}T_1) = \{s^{-1}w\} \end{aligned}$$

so $s^{-1} \in S$.

Now let $g \in G$ and consider gv_0. There is a unique path in T from gv_0 to v_0, passing through tiles

$$T_{g_n}, T_{g_{n-1}}, \ldots, T_{g_1}, T_{g_0}$$

We claim that $g_{i-1}^{-1}g_i \in S$. The above path travels from $T_{g_{i+1}}$ to T_{g_i} without passing through a middle tile, so

$$T_{g_{i+1}} \cap T_{g_i} \neq \varnothing$$

Applying g_i^{-1}, we get

$$g_i^{-1}T_{g_{i+1}} \cap T_1 \neq \varnothing \implies T_{g_i^{-1}g_{i+1}} \cap T_1 \neq \varnothing$$

so $g_i^{-1}g_{i+1} \in S$, and we write it as s_i. Then, we have that

$$\begin{aligned} g &= g_n \\ &= g_{n-1}g_{n-1}^{-1}g_n \\ &\vdots \\ &= g_1^{-1}g_1g_2^{-1}g_2 \cdots g_{n-1}^{-1}g_n \\ &= s_1s_2 \cdots s_n \end{aligned}$$

so g may be written as a product of elements of S, meaning S generates G. □

With all these tools now in place, the last thing we need to do is show that there are no relations between elements of S.

Proof of Theorem 2.17. By Lemma 2.19 we have a G-tiling $\{T_g\}$, which we know generates G by Lemma 2.23. We claim that every $g \in G$ may be written as a unique, reduced product of elements of S. Let

$$g = s_1s_2 \cdots s_k$$

with $s_i \neq s_{i+1}^{-1}$. We claim there is a unique path from gv_0 to v_0 that passes through, in order, the tiles

$$T_g = T_{s_1s_2\cdots s_k}, T_{s_1s_2\cdots s_{k-1}}, \ldots, T_{s_1}, T_1$$

As $T_{s_1} \cap T_1 = \{w_1\}$ for some $w_1 \in V(T)$, $T_{s_1} \cup T_1$ is a tree, where $v_0 \in V(T_1)$ and $s_1v_0 \in T_{s_1}$. Thus, there is a unique path from s_1v_0 to v_0 in $T_{s_1} \cup T_1$. Similarly, we have

$$T_{s_1s_2} \cap T_{s_1} = s_1^{-1}(T_{s_2} \cap T_1) = \{s_1^{-1}w_2\}$$

for some $w_2 \in V(T)$, so $T_{s_1s_2} \cup T_{s_1}$ is a tree. Again, this means there is a unique path from s_1v_0 to $s_1s_2v_0$ in $T_{s_1s_2} \cup T_1$. If we continue this process inductively, we get the desired unique path. This path completely deterimines the word $s_1s_2 \cdots s_k$ representing g, and thus g can only be written as this freely reduced word in S. This tells us that S has no relations, and thus G is free, as desired. □

Corollary 2.24 (Nielsen-Schreier Theorem)**.** *Ever subgroup of a free group is also a free group.*

Proof. An action of a group G on a tree trivially restricts to an action of $H \leq G$ on the tree. As $1 \in H$, if the action of G is free, so too must the action of H. □

References

[Ady55] S. I. Adyan. "Algorithmic unsolvability of problems of recognition of certain properties of groups". In: *Doklady Akademii Nauk SSSR* 103 (1955).

[Cay78] Arthur Cayley. "Desiderata and suggestions: No. 2. The Theory of groups: graphical representation". In: *American Journal of Mathematics* 1.2 (1878).

[Dyc82] Walther von Dyck. "Gruppentheoretische Studien". In: *Mathematische Annalen* 20.3 (1882).

[Fuc80] Lazarus Fuchs. "Ueber eine Klasse von Funktionen mehrerer Variablen, welche durch Umkehrung der Integrale von Lösungen der linearen Differentialgleichungen mit rationalen Coeffizienten entstehen". In: *J. Reine Angew. Math* 89 (1880).

[Har00] Pierre de la Harpe. *Topics in Geometric Group Theory*. University of Chicago Press, 2000.

[Mar17] Matt Clay & Dan Margalit, ed. *Office Hours with a Geometric Group Theorist*. Princeton University Press, 2017.

[Nie21] Jakob Nielsen. "On calculation with noncommutative factors and its application to group theory". In: *The Mathematical Scientist* 6.2 (1921).

[Nie24] Jakob Nielsen. "Die Isomorphismengruppe der freien Gruppen". In: *Mathematische Annalen* 91.3 (1924).

[Rab58] Michael O. Rabin. "Recursive unsolvability of group theoretic problems". In: *Annals of Mathematics* 67.2 (1958).

[Sab58] Gert Sabidussi. "On a class of fixed-point-free-graphs". In: *Proceedings of the American Mathematical Society* 9.5 (1958).

[Sch28] Otto Schreier. "Die Untergruppen der freien Gruppen". In: *1* 5.Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg (1928).

[Ser80] J.P. Serre. *Trees*. Ed. by J. Stillwell. Springer Verlag, 1980.

[Tit72] Jacques Tits. "Free subgroups in linear groups". In: *Journal of Algebra* 20.2 (1972).